

3.2) a) ON base: $(\bar{u}_1, \bar{u}_2, \bar{u}_3)$

$$(\bar{v}_1, \bar{v}_2, \bar{v}_3) = (1, x, x^2)$$

$$\bar{u}_1 = \frac{\bar{v}_1}{|\bar{v}_1|} = \frac{1}{\sqrt{\int_1^5 1^2 dx}} = 2$$

$$\bar{w}_2 = \langle \bar{u}_1, \bar{v}_2 \rangle \bar{u}_1 = \left(\int_1^5 2x dx \right) \cdot 2 = 2\sqrt{6} \cdot 2 = 4\sqrt{6}$$

$$\bar{v}_2 - \bar{w}_2 = x - 4\sqrt{6}$$

$$\bar{u}_2 = \frac{1}{|x - 4\sqrt{6}|} (x - 4\sqrt{6}) = \frac{1}{\sqrt{\int_1^5 (x - 4\sqrt{6})^2 dx}} \cdot (x - 4\sqrt{6}) \approx 0,07x + 0,7$$

$$\bar{w}_3 = \langle \bar{u}_1, \bar{v}_3 \rangle \bar{u}_1 + \langle \bar{u}_2, \bar{v}_3 \rangle \bar{u}_2$$

$$= \left(\int_1^5 2x^2 dx \right) \cdot 2 + \left(\int_1^5 \frac{(x + 4\sqrt{6})x^2}{\sqrt{\frac{1276}{3} - 96\sqrt{6}}} dx \right) \cdot \frac{x + 4\sqrt{6}}{\sqrt{\frac{1276}{3} - 96\sqrt{6}}} \approx 2,95x + 194,2$$

$$\bar{v}_3 - \bar{w}_3 = x^2 - 2,95x + 194,2$$

$$\bar{u}_3 = \frac{1}{|x^2 - 2,95x + 194,2|} (x^2 - 2,95x + 194,2) \approx \frac{1}{385,5} (x^2 - 2,95x - \frac{194,2}{2})$$

$$\text{ON base: } \left(\underbrace{2}_{\tilde{u}_1}, \underbrace{0,07x+0,7}_{\tilde{u}_2}, \underbrace{\frac{1}{385,5} (x^2 - 2,95x - 194,2)}_{\tilde{u}_3} \right)$$

$$b) \tilde{w} = \frac{1}{385,5} \left(\langle 2, \sin x \rangle \cdot 2 + \langle 0,07x+0,7, \sin x \rangle (0,07x+0,7) + \langle x^2 - 2,95x - 194,2, \sin x \rangle \cdot \frac{1}{385,5} (x^2 - 2,95x - 194,2) \right)$$

$$\tilde{w} = 2 \cdot \int_1^5 2 \cdot \sin x dx + (0,07x+0,7) \cdot \int_1^5 \sin x (0,07x+0,7) dx + \tilde{u}_3 \int_1^5 \sin x \cdot \tilde{u}_3 dx$$

$$\tilde{w} \approx 1,03 + \cancel{0,0008 (0,07x+0,7)} 5,6 \cdot 10^{-4} x - 5,6 \cdot 10^{-3} + 4 \cdot 10^{-4} x^2 + 1,18 \cdot 10^{-3} x + 0,08$$

$$\tilde{w} \approx \underline{\underline{-4 \cdot 10^{-4} x^2 + 6,2 \cdot 10^{-4} x + 1,1}}$$

$$c) \left| \sin x - \underbrace{(-4 \cdot 10^{-4} x^2 + 6,2 \cdot 10^{-4} x + 1,1)}_a \right| = \sqrt{\langle \sin x - a, \sin x - a \rangle} = \sqrt{\int_1^5 (\sin x - a)^2 dx}$$

$$\approx \underline{\underline{2,57}}$$